

Electronic Equivalent Time Exchange Protocol (EEP)

EEP Specification

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0.1 Document Information	

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1 Revision History

Version	Date	Description
1.2	2026	Removed DELETE processing and associated validation rules. DELETE requests may still be recognized for backward compatibility but are no longer supported for security and data integrity reasons.
1.1	2026	Added DELETE operation and associated validation rules.
1.0	2026	Initial public release.

2 Status of this Document

This document defines Version 1.2 of the Electronic Equivalent Time Exchange Protocol (EEP).

This document is the official specification of Version 1.2 of the Electronic Equivalent Time Exchange Protocol (EEP).

Version 1.2 is fully backward compatible with Version 1.0 for all operations defined by Version 1.0.

3 Introduction

3.1 Purpose

The Electronic Equivalent Time Exchange Protocol (EEP) defines a simple HTTP/JSON protocol for exchanging the data required to calculate an Equivalent Electronic Time (EET).

The protocol enables timing systems and EET calculation software to exchange a standardized JSON document through HTTP requests. It reduces manual data entry while preserving user control over the calculation process.

EEP specifies:

- the JSON document structure;
- the HTTP message exchange;
- the validation rules required for interoperability.

The protocol is independent of any timing software or EET calculation implementation. It defines only the exchange of data required to perform an EET calculation.

The protocol does not define the EET calculation itself.

EEP is intended to serve as an implementation-independent exchange protocol for Equivalent Electronic Time (EET) calculations.

The protocol is openly documented in order to encourage interoperability between timing systems and EET software.

3.2 Design Goals

EEP has been designed with the following objectives:

- simple implementation;
- minimal manual data entry;
- backward-compatible extensibility.

4 Workflow

The EEP workflow consists of one mandatory exchange and one optional exchange.

The Initial Request creates a new EET calculation and provides the electronic timing data available from the timing system.

The Manual Time (MT) may then be provided either through the EET Calculator user interface or by an Optional Secondary Request.

The user reviews the calculation data before validating the final Equivalent Electronic Time (EET).

5 HTTP Interface

EEP uses HTTP POST requests with JSON payloads.

The endpoint URL is implementation-specific.

All requests and responses SHALL use the following content type:

Content-Type: `application/json`

The protocol defines four message types.

Message	Purpose
Initial Request	Create a new EET calculation.
Initial Response	Return the assigned <code>calculation_id</code> .
Optional Secondary Request	Provide the previously missing Manual Time.
Secondary Response	Confirm the successful update.

A successful Initial Response assigns a unique `calculation_id` that SHALL be used in all subsequent requests related to the calculation.

6 EEP Document

The EEP document is a JSON object exchanged between a timing system and an EET Calculator.

It consists of a root object containing race information and a list of competitors.

Optional members provide additional information intended to improve the presentation of the EET calculation result.

Member ordering has no semantic meaning.

6.1 Root Object

The root object contains the calculation identifier, processing mode and race information. Depending on the request type, it may also contain a competitor list.

Member	Type	Required	Description
<code>calculation_id</code>	string	Yes	Unique calculation identifier. Empty in the Initial Request.
<code>race</code>	object	Yes	Race information.
<code>competitors</code>	array	Yes	List of competitors. Initial Requests and Optional Secondary Requests SHALL contain exactly 11 competitors.
<code>mode</code>	string	No	Processing mode. Recognized values are "" (or omitted) and <code>TEST</code> .

Recognized values for `mode` are:

Value	Meaning
"" or omitted	Normal calculation.
<code>TEST</code>	Test calculation.

6.1.1 Root Object Example

```
{
  "calculation_id": "",
  "mode": "TEST",
  "race": {
    ...
  },
}
```

```

    "competitors": [
        ...
    ]
}

```

6.1.2 Race Object

The race object identifies the competition associated with the calculation and specifies the competitor for whom an Equivalent Electronic Time (EET) is requested.

Member	Type	Required	Description
season	string	Yes	Competition season.
codex	string	Yes	Competition codex.
run	string	Yes	Competition run number.
missing_impulse	string	Yes	Missing timing impulse (START or FINISH).
eet_bib	string	Yes	Bib number of the competitor requiring an EET.
name	string	No	Competition name.
discipline	string	No	Competition discipline.
date	string	No	Competition date.
location	string	No	Competition location.

6.1.2.1 Race Object Example

```

{
  "season": "2026",
  "codex": "1234",
  "run": "1",
  "missing_impulse": "START",
  "eet_bib": "14",
  "name": "National Championships",
  "discipline": "SL",
  "date": "2026-02-14",
  "location": "Val d'Isère"
}

```

6.2 Competitor Object

Each element of the `competitors` array represents one competitor included in the EET calculation.

The competitor whose electronic timing impulse is missing is identified by an empty `et_tod` value in the Initial Request.

Member	Type	Required	Description
bib	string	Yes	Competitor bib number.
et_tod	string	Yes	Electronic Time expressed as Time of Day.
name	string	No	Competitor first name.
surname	string	No	Competitor surname.
nation	string	No	Competitor nation.
club	string	No	Competitor club.

6.2.1 Competitor Object Example

```

{
  "bib": "14",
  "et_tod": "",
  "name": "John",
  "surname": "Smith",
  "nation": "FRA",
  "club": "SC Val d'Isère"
}

```

7 JSON Object Hierarchy

The following hierarchy summarizes the structure of the EEP JSON document.

```
Root Object
|
+-- calculation_id
|
+-- mode
|
+-- race
|   |
|   +-- season
|   +-- codex
|   +-- run
|   +-- missing_impulse
|   +-- eet_bib
|   +-- name
|   +-- discipline
|   +-- date
|   +-- location
|
+-- competitors[]
    |
    +-- bib
    +-- et_tod
    +-- name
    +-- surname
    +-- nation
    +-- club
```

8 Validation Rules

The following rules define the minimum validation requirements for protocol interoperability.

Implementations MAY perform additional business validation without affecting interoperability.

Rule	Description
VR-01	For Initial Requests and Optional Secondary Requests, the root object SHALL contain <code>calculation_id</code> , <code>race</code> and <code>competitors</code> .
VR-02	The race object SHALL contain <code>season</code> , <code>codex</code> , <code>run</code> , <code>missing_impulse</code> and <code>eet_bib</code>
VR-03	Each competitor SHALL contain <code>bib</code> and <code>et_tod</code> .
VR-04	The <code>competitors</code> array SHALL contain exactly eleven competitors.
VR-05	In an Initial Request, exactly one competitor SHALL have an empty <code>et_tod</code> value.
VR-06	In an Optional Secondary Request, no <code>et_tod</code> value SHALL be empty.
VR-07	Every non-empty <code>et_tod</code> value SHALL contain at least three decimal places.
VR-08	<code>season</code> SHALL consist of exactly four digits.
VR-09	<code>missing_impulse</code> SHALL be either <code>START</code> or <code>FINISH</code> .
VR-10	Unknown members SHALL be ignored to preserve forward compatibility.

8.1 Backward Compatibility — DELETE Requests

For backward compatibility with Version 1.1, an EEP Version 1.2 implementation MAY recognize a request with `mode` set to `DELETE`.

Such a request SHALL NOT be processed and SHALL NOT result in the deletion of any stored calculation.

The implementation SHALL return an error with the code `UNSUPPORTED_MODE`.

The `DELETE` mode is no longer a supported EEP operation in Version 1.2.

9 Protocol Examples

The following examples illustrate the HTTP messages exchanged between a timing system and an EET Calculator.

The Optional Secondary Request differs from the Initial Request only by:

- the assigned `calculation_id`;
- the previously missing `et_tod` value.

For clarity, the examples include optional members. Implementations MAY omit any optional member.

Optional competitor members are shown only for the competitor whose electronic timing impulse is missing.

9.1 Initial Request

POST /eet HTTP/1.2

Content-Type: application/json

```
{
  "calculation_id": "",
  "mode": "TEST",
  "race": {
    "season": "2026",
    "codex": "1234",
    "run": "1",
    "missing_impulse": "START",
    "eet_bib": "14",
    "name": "National Championships",
    "discipline": "SL",
    "date": "2026-02-14",
    "location": "Val d'Isère"
  },
  "competitors": [
    {
      "bib": "11",
      "et_tod": "10:16:12.481"
    },
    {
      "bib": "12",
      "et_tod": "10:16:24.106"
    },
    {
      "bib": "13",
      "et_tod": "10:16:32.547"
    },
    {
      "bib": "14",
      "et_tod": "",
      "name": "John",
      "surname": "Smith",
      "nation": "FRA",
      "club": "SC Val d'Isère"
    },
    {
      "bib": "15",
      "et_tod": "10:16:51.234"
    },
    {
      "bib": "16",
      "et_tod": "10:17:03.118"
    },
    {
      "bib": "17",
      "et_tod": "10:17:14.904"
    }
  ]
}
```

```

    },
    {
      "bib": "18",
      "et_tod": "10:17:26.431"
    },
    {
      "bib": "19",
      "et_tod": "10:17:39.055"
    },
    {
      "bib": "20",
      "et_tod": "10:17:52.601"
    },
    {
      "bib": "21",
      "et_tod": "10:18:05.227"
    }
  ]
}

```

9.2 Initial Response

HTTP/1.2 200 OK
Content-Type: application/json

```

{
  "status": "ok",
  "calculation_id": "A7F4K2"
}

```

9.3 Optional Secondary Request

POST /eet HTTP/1.2
Content-Type: application/json

```

{
  "calculation_id": "A7F4K2",
  "mode": "TEST",
  "race": {
    "season": "2026",
    "codex": "1234",
    "run": "1",
    "missing_impulse": "START",
    "eet_bib": "14",
  },
  "competitors": [
    {
      "bib": "11",
      "et_tod": "10:16:12.481"
    },
    {
      "bib": "12",
      "et_tod": "10:16:24.106"
    },
    {
      "bib": "13",
      "et_tod": "10:16:32.547"
    },
    {
      "bib": "14",
      "et_tod": "10:16:40.216",
      "name": "John",
      "surname": "Smith",
    }
  ]
}

```



```

    "nation": "FRA",
    "club": "SC Val d'Isère"
  },
  {
    "bib": "15",
    "et_tod": "10:16:51.234"
  },
  {
    "bib": "16",
    "et_tod": "10:17:03.118"
  },
  {
    "bib": "17",
    "et_tod": "10:17:14.904"
  },
  {
    "bib": "18",
    "et_tod": "10:17:26.431"
  },
  {
    "bib": "19",
    "et_tod": "10:17:39.055"
  },
  {
    "bib": "20",
    "et_tod": "10:17:52.601"
  },
  {
    "bib": "21",
    "et_tod": "10:18:05.227"
  }
]
}

```

9.4 Secondary Response

HTTP/1.2 200 OK
Content-Type: application/json

```

{
  "status": "ok",
  "calculation_id": "A7F4K2"
}

```

Implementation Note

During processing of an Optional Secondary Request, implementations SHOULD update only the previously missing `et_tod` value. All other document members SHOULD be ignored.

10 Appendix A — JSON Naming Conventions

EEP follows common JSON naming conventions.

Convention	Rule
Member names	lowercase
Word separator	underscore (<code>_</code>)
Arrays	plural nouns
Boolean values	JSON <code>true</code> / <code>false</code>
Null values	JSON <code>null</code>
Date format	ISO 8601 (YYYY-MM-DD)
Time of Day	HH:MM:SS.sss (minimum three decimal places)

These conventions are recommendations intended to improve readability and consistency across implementations.

11 Appendix B — Error Codes

When a request cannot be processed, the EET Calculator returns an HTTP 400 (Bad Request) response with the following JSON structure.

```
{
  "status": "error",
  "messages": [
    "<ERROR_CODE>"
  ]
}
```

The `messages` member is an array, allowing multiple validation errors to be returned in a single response.

The following error codes are defined by the EEP protocol.

Error Code	Description
INVALID_COMPETITOR_COUNT	The number of competitors is outside the allowed range.
DUPLICATE_BIB	The competitors list contains duplicate bib numbers.
INVALID_ET_COUNT	The number of electronic times does not satisfy the protocol requirements.
INVALID_ET_PRECISION	The electronic time precision is outside the supported range.
INVALID_MT	One or more manual times are invalid.
UNKNOWN_CALCULATION_ID	The specified <code>calculation_id</code> does not correspond to any stored calculation.
BUSINESS_KEY_MISMATCH	The supplied business key (<code>season</code> , <code>codex</code> , <code>run</code> , <code>eet_bib</code> , <code>missing_impulse</code>) does not match the calculation identified by <code>calculation_id</code> .
UNSUPPORTED_MODE	The requested processing mode is not supported by this version of the EEP implementation.

Error codes are stable protocol identifiers intended for machine processing and **must not be translated**. Client applications may display localized messages corresponding to these codes.

12 Copyright and Permissions

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Modified versions of this specification shall clearly identify the modifications and shall not be presented as the official EEP Specification.

The name “EEP Specification” refers exclusively to the official specification published by the author.